

Module 1: The Scientific Method & Basic Chemistry

Comprehensive Identification & Multiple-Choice Reviewer — 152 Items

Name: _____

Date: _____ Score: _____

INSTRUCTIONS: Read each statement carefully. Identify the correct term or concept described in the blank (____). Choose the letter of the correct answer from the choices provided (a, b, c, or d) and write it on the blank provided. An exhaustive Answer Key with detailed rationales is provided at the end of this document.

PART I: IDENTIFICATION QUESTIONNAIRE (152 ITEMS)

- ____ 1. A systematic way of learning about the natural world by applying rules of logic to the way information is gathered and conclusions are drawn is the ____.
- a.) natural law b.) inductive philosophy c.) scientific method d.) anecdotal observation
- ____ 2. In the scientific method, Phase 1 begins when careful observation prompts a ____ that is grounded in existing scientific knowledge.
- a.) proof b.) question c.) dogma d.) conclusion
- ____ 3. According to the scientific method, questions prompted by observations should be based on existing knowledge gathered through observation of the ____.
- a.) virtual model b.) speculation c.) environment d.) mathematical theory
- ____ 4. A testable possible explanation formulated as a tentative answer to an observed question is a ____.
- a.) hypothesis b.) variable c.) scientific law d.) paradigm
- ____ 5. To be methodologically sound and scientifically valid, a hypothesis must always be phrased as a ____, rather than as an interrogative question.
- a.) paradox b.) question c.) statement d.) metaphor
- ____ 6. In scientific inquiry, the proposition 'Daily exposure to the sun leads to increased levels of happiness' is a classic example of a ____.
- a.) scientific law b.) conclusion c.) hypothesis d.) control group
- ____ 7. A fundamental characteristic of any valid hypothesis is that it must be testable, which means that it has the potential to be proven ____.
- a.) unquestionable b.) infallible c.) absolute d.) false
- ____ 8. In the philosophy of science, while empirical experimentation can definitively prove a hypothesis false, it can never be proven as ____.
- a.) true b.) false c.) testable d.) provisional
- ____ 9. In Phase 3 of the scientific method, researchers make a ____ based on the hypothesis, which guides the experimental design.
- a.) postulate b.) prediction c.) conclusion d.) axiom
- ____ 10. In experimental methodology, the formulated prediction will determine the experiment or observations that are necessary to ____ the hypothesis.
- a.) crystallize b.) discard c.) test d.) canonize
- ____ 11. An experiment that tests a hypothesis by comparing an experimental cohort with a reference cohort under identical conditions is a ____ experiment.
- a.) spontaneous b.) speculative c.) controlled d.) subjective
- ____ 12. In a controlled experiment, the two cohorts compared to evaluate the hypothesis are the experimental group and the ____ group.

a.) alternate b.) variable c.) dependent d.) control

_____ 13. In a controlled experiment, the control group and the experimental group differ only on a single factor known as a _____.

a.) hypothesis b.) variable c.) constant d.) placebo

_____ 14. In a controlled experiment, the specific factor whose biological effect is being revealed is called the _____.

a.) control b.) constant c.) variable d.) sample size

_____ 15. In Phase 4 of the scientific method, researchers draw a _____ based on the rigorous interpretation of experimental data.

a.) variable b.) conclusion c.) prediction d.) hypothesis

_____ 16. In the scientific method, the final conclusion reached by researchers is based directly on the interpretation of _____.

a.) consensus b.) data c.) intuition d.) tradition

_____ 17. To ensure that experimental results and differences between groups are not due to chance alone, researchers require _____.

a.) anecdotal consensus b.) peer agreement c.) subjective confidence d.) statistical significance

_____ 18. Anything that possesses mass and occupies physical space is defined in chemistry as _____.

a.) energy b.) radiation c.) matter d.) vacuum

_____ 19. Matter is composed of fundamental structural units called _____, which cannot be broken down into simpler substances by ordinary chemical means.

a.) cells b.) compounds c.) atoms d.) macromolecules

_____ 20. Although indivisible by chemical means, atoms are composed of smaller subatomic particles, namely protons, electrons, and _____.

a.) neutrons b.) positrons c.) quarks d.) photons

_____ 21. The subatomic particle located within the atomic nucleus that possesses a positive electrical charge is the _____.

a.) neutron b.) proton c.) electron d.) neutrino

_____ 22. The subatomic particle located in the nucleus that carries no electrical charge (is neutral) is the _____.

a.) electron b.) ion c.) proton d.) neutron

_____ 23. The subatomic particle that orbits the nucleus and carries a negative electrical charge is the _____.

a.) electron b.) positron c.) proton d.) neutron

_____ 24. If a pure form of matter contains only one kind of atom, it is identified as an _____.

a.) mixture b.) isotope c.) element d.) compound

_____ 25. In the physical universe and periodic table, there are more than _____ known chemical elements.

a.) 100 b.) 500 c.) 20 d.) 50

_____ 26. Although there are more than 100 known elements, only about _____ elements are found naturally in the human body.

a.) 4 b.) 20 c.) 92 d.) 50

_____ 27. The vast majority of human body tissue is composed mostly of four primary elements: carbon, oxygen, nitrogen, and _____.

a.) calcium b.) potassium c.) phosphorus d.) hydrogen

_____ 28. In standard chemical nomenclature, chemical elements are abbreviated by a _____ symbol.

a.) three-letter b.) Greek letter c.) numerical d.) one or two-letter

_____ 29. The standard metric unit used to quantify and measure the atomic weight of an element is the _____.

a.) joule b.) atomic mass unit c.) gram per liter d.) angstrom

_____ 30. The atomic weight of any given chemical element is primarily influenced by the total number of protons and _____.

a.) neutrons b.) electrons c.) orbitals d.) photons

- _____ 31. Atoms of the same element that contain the same number of protons but differ in their number of neutrons are called _____.
a.) allotropes b.) isomers c.) isotopes d.) monomers
- _____ 32. Unstable isotopes that spontaneously break down over time and emit radiation or subatomic particles are called _____.
a.) stable isotopes b.) ions c.) electrolytes d.) radioisotopes
- _____ 33. When radioisotopes decay and break down, they release particles or _____.
a.) radiation b.) nitrogen gas c.) ozone d.) glucose
- _____ 34. When two or more different elements chemically combine, they form a new substance called a _____, which has totally different characteristics from the parent elements.
a.) compound b.) mixture c.) element d.) isotope
- _____ 35. Atoms within chemical substances are held together through attractive forces known as chemical _____.
a.) charges b.) bonds c.) gradients d.) currents
- _____ 36. The chemical bonds that hold atoms together in biological molecules are broadly categorized into ionic bonds and _____ bonds.
a.) hydrophobic b.) peptide c.) metallic d.) covalent
- _____ 37. Chemical bonds that are formed when two or more atoms share electrons in their outer cells (outer shells) are _____ bonds.
a.) electrostatic b.) covalent c.) hydrogen d.) ionic
- _____ 38. Covalent bonds are specifically formed when two or more atoms share electrons located in their _____.
a.) outer shells b.) proton core c.) inner nucleus d.) neutron cloud
- _____ 39. Chemical bonds formed from the transfer of valence electrons and the resulting mutual attraction of oppositely charged ions are _____ bonds.
a.) ionic b.) hydrogen c.) covalent d.) disulfide
- _____ 40. Ionic bonds are fundamentally driven by the mutual attraction between _____ charged ions.
a.) neutrally b.) identically c.) covalently d.) oppositely
- _____ 41. Because water is capable of dissolving a vast variety of chemical substances, it is widely termed the universal biological _____.
a.) solvent b.) solute c.) buffer d.) polymer
- _____ 42. Water can absorb large amounts of heat energy with minimal changes in its own temperature because it features a high _____.
a.) heat capacity b.) viscosity c.) density d.) vapor pressure
- _____ 43. The property of water that enables perspiration to serve as an exceptionally effective cooling mechanism is its high heat of _____.
a.) fusion b.) condensation c.) crystallization d.) vaporization
- _____ 44. Water's distinctive qualities, such as its high heat capacity and solvent power, are due to its ability to form hydrogen bonds and the molecule's _____.
a.) polarity b.) nonpolar nature c.) neutral charge d.) linear geometry
- _____ 45. In water chemistry, polarity refers specifically to the presence of _____ charges within the water molecule.
a.) ionic b.) neutral c.) partial d.) complete
- _____ 46. The attraction between slightly positive and slightly negative charges of adjacent water molecules forms a _____ bond.
a.) covalent b.) peptide c.) hydrogen d.) ionic
- _____ 47. In a hydrogen bond between water molecules, the attraction occurs between slightly _____ and slightly negative charges.
a.) positive b.) covalent c.) neutral d.) identical
- _____ 48. Large biological molecules formed by thousands of atoms linked together into long chains are called _____.

a.) macromolecules b.) coenzymes c.) microelements d.) micelles

_____ 49. If macromolecules consist of small, repeating molecular subunits linked in a chain, they are identified as _____.

a.) electrolytes b.) monomers c.) dimers d.) polymers

_____ 50. The individual, small molecular building blocks that link together to form polymers are referred to as _____.

a.) complexes b.) polymers c.) isomers d.) monomers

_____ 51. Chemical reactions that join monomers together to form polymers by eliminating water molecules are _____ reactions.

a.) oxidation b.) hydrolysis c.) carboxylation d.) dehydration

_____ 52. Chemical reactions that break polymers down into individual monomers by consuming water molecules are _____ reactions.

a.) dehydration b.) condensation c.) hydrolysis d.) polymerization

_____ 53. In cellular biochemistry, while dehydration reactions form polymers, _____ reactions break polymers down.

a.) synthesis b.) hydrolysis c.) dehydration d.) anabolic

_____ 54. The four major types of biomolecules essential to life are carbohydrates, proteins, nucleic acids, and _____.

a.) vitamins b.) minerals c.) electrolytes d.) fats

_____ 55. Biomolecules made entirely of carbon, hydrogen, and oxygen that primarily provide fuel or energy for the human body are _____.

a.) proteins b.) nucleic acids c.) carbohydrates d.) lipids

_____ 56. Common dietary examples of carbohydrates include starches and _____.

a.) amino acids b.) triglycerides c.) steroids d.) sugars

_____ 57. Carbohydrates are classified into three major hierarchical groups: monosaccharides, polysaccharides, and _____.

a.) oligosaccharides b.) fatty acids c.) polypeptides d.) phospholipids

_____ 58. The simplest carbohydrate units, commonly referred to as simple sugars, are _____.

a.) disaccharides b.) monosaccharides c.) oligosaccharides d.) polysaccharides

_____ 59. Monosaccharides are categorized by the number of carbon atoms in their ring, which ranges from three to _____ carbon atoms.

a.) twelve b.) seven c.) four d.) ten

_____ 60. Monosaccharides containing six carbon atoms, such as glucose, fructose, and galactose, are classified as _____.

a.) hexoses b.) heptoses c.) pentoses d.) trioses

_____ 61. The hexose monosaccharide noted for being rich in carbohydrates and circulating as the primary fuel for human cells is _____.

a.) ribose b.) cellulose c.) glucose d.) sucrose

_____ 62. The hexose monosaccharide naturally seen and found in abundance in fruits and vegetables is _____.

a.) fructose b.) galactose c.) maltose d.) glycogen

_____ 63. The hexose monosaccharide commonly seen in dairy products is _____.

a.) fructose b.) starch c.) galactose d.) ribose

_____ 64. The five-carbon monosaccharide that is classified as a pentose sugar is _____.

a.) glucose b.) galactose c.) fructose d.) ribose

_____ 65. A five-carbon simple sugar, such as ribose, is scientifically termed a _____.

a.) tetrose b.) pentose c.) hexose d.) triose

_____ 66. Carbohydrate chains composed of a few monosaccharides joined together by dehydration synthesis are called _____.

a.) oligosaccharides b.) lipids c.) polysaccharides d.) monosaccharides

- _____ 67. Oligosaccharides are categorized and classified based on the number of _____ present per chain.
a.) electrons b.) subunits c.) peptide links d.) double bonds
- _____ 68. Double sugars that form when two monosaccharides covalently bond to each other through dehydration synthesis are _____.
a.) hexoses b.) pentoses c.) disaccharides d.) polysaccharides
- _____ 69. A classic example of a disaccharide, commonly known as household table sugar, is _____.
a.) cellulose b.) glycogen c.) glucose d.) sucrose
- _____ 70. Complex carbohydrates wherein monosaccharides join together in longer and more extensive chains are _____.
a.) polysaccharides b.) disaccharides c.) oligosaccharides d.) hexoses
- _____ 71. Polysaccharides are categorized based on their biological function into two main types: structural and _____.
a.) amphipathic b.) storage c.) transport d.) catalytic
- _____ 72. Polysaccharides that function to store energy for future physiological use are classified as _____ polysaccharides.
a.) structural b.) genetic c.) hormonal d.) storage
- _____ 73. The primary storage polysaccharide found in plants is _____.
a.) cellulose b.) starch c.) chitin d.) glycogen
- _____ 74. The primary storage polysaccharide found in animals, stored in liver and muscle tissue, is _____.
a.) glycogen b.) sucrose c.) starch d.) cellulose
- _____ 75. Polysaccharides that provide physical strength and mechanical support in organisms are classified as _____ polysaccharides.
a.) structural b.) nutritional c.) enzymatic d.) storage
- _____ 76. The structural polysaccharide that provides strength in organisms and is seen in materials like bamboo, jute, and cotton is _____.
a.) glycogen b.) cellulose c.) amylose d.) starch
- _____ 77. Plant materials such as bamboo, jute, and cotton are highlighted as prominent examples containing the structural carbohydrate _____.
a.) galactose b.) cellulose c.) sucrose d.) glycogen
- _____ 78. Organic biomolecules that do not dissolve in water due to their nonpolar nature are _____.
a.) proteins b.) lipids c.) nucleic acids d.) carbohydrates
- _____ 79. The three types of lipids that are critically important to human health are phospholipids, steroids, and _____.
a.) amino acids b.) triglycerides c.) polysaccharides d.) peptides
- _____ 80. A triglyceride molecule is comprised of one molecule of glycerol chemically bonded to three _____.
a.) fatty acids b.) amino acids c.) nucleotides d.) phosphate groups
- _____ 81. Fatty acids are chains of carbon atoms bonded to hydrogens that feature the functional group _____ at one end of the chain.
a.) COOH b.) OH c.) NH₂ d.) PO₄
- _____ 82. In fatty acid chemical structure, chains of carbon atoms are covalently bonded to _____.
a.) phosphates b.) sulfurs c.) nitrogens d.) hydrogens
- _____ 83. Fatty acids are categorized into saturated and unsaturated based on the presence of _____.
a.) phosphate groups b.) ring structures c.) double bonds d.) amino groups
- _____ 84. If no double bonds are present between carbon atoms in a fatty acid chain, it is called a _____ fatty acid.
a.) polyunsaturated b.) trans c.) saturated d.) unsaturated
- _____ 85. If there is at least one double bond present in a fatty acid chain, it is classified as an _____ fatty acid.
a.) unsaturated b.) saturated c.) solid d.) hydrogenated
- _____ 86. Common dietary examples of triglycerides include fats and _____.

a.) vitamins b.) starches c.) sugars d.) oils

_____ **87.** Triglycerides serve as important storage compounds in animals because per gram, they store _____ the energy of carbohydrates.

a.) twice b.) ten times c.) half d.) the same

_____ **88.** The most common type of fat found in the human body is the _____.

a.) steroid b.) triglyceride c.) phospholipid d.) wax

_____ **89.** Triglycerides in the human body come from dietary foods you eat, especially butter, other fats, and _____.

a.) oils b.) grains c.) fruits d.) vegetables

_____ **90.** A phospholipid molecule is comprised of a glycerol molecule bonded to a negatively charged phosphate group and _____ fatty acids.

a.) one b.) two c.) four d.) three

_____ **91.** Attached to the negatively charged phosphate group of a phospholipid is often a small molecule known as the _____ group.

a.) steroid b.) variable c.) peptide d.) monomer

_____ **92.** Molecules that possess both hydrophobic (water-repelling) and hydrophilic (water-loving) regions are described as _____.

a.) amphipathic b.) nonpolar c.) hydrophilic d.) hydrophobic

_____ **93.** Because phospholipids are amphipathic, they function as fundamental structural components of the _____ membrane.

a.) nuclear pore b.) cell wall c.) plasma d.) ribosome

_____ **94.** Steroids are a distinctive class of lipids whose core molecular structure is composed of _____ carbon rings.

a.) three b.) two c.) six d.) four

_____ **95.** In steroid molecular architecture, specific parts of the four carbon rings are attached to _____ molecules.

a.) glycerol b.) nucleotide c.) polypeptide d.) variable

_____ **96.** The essential steroid that is an integral component of the plasma membrane and serves as the precursor for other steroid hormones is _____.

a.) insulin b.) cholesterol c.) triglyceride d.) glycogen

_____ **97.** Cholesterol serves as the biochemical foundation for synthesizing essential steroid hormones such as testosterone and _____.

a.) adrenaline b.) thyroxine c.) insulin d.) estrogen

_____ **98.** Polymers comprised from a selection of 20 different building blocks linked together to form chains are _____.

a.) polysaccharides b.) lipids c.) proteins d.) nucleic acids

_____ **99.** The fundamental monomeric building block of all proteins is the _____.

a.) fatty acid b.) amino acid c.) nucleotide d.) monosaccharide

_____ **100.** Every amino acid is structured around a central _____ atom bound to a hydrogen atom, an amino group, a carboxyl group, and a side chain.

a.) carbon b.) nitrogen c.) phosphorus d.) oxygen

_____ **101.** In an amino acid, the functional group containing nitrogen bound to hydrogens (-NH₂) is the _____ group.

a.) carboxyl b.) hydroxyl c.) amino d.) phosphate

_____ **102.** In an amino acid, the acidic functional group bound to the central carbon is the _____ group.

a.) methyl b.) phosphate c.) amino d.) carboxyl

_____ **103.** The specific part of an amino acid molecule that differentiates it from all other amino acids is the _____.

a.) side chain b.) central carbon c.) amino group d.) carboxyl group

_____ **104.** Amino acids are classified into essential and non-essential based on the ability of the _____ to produce it.

a.) climate b.) body c.) diet d.) bacteria

- _____ **105.** Amino acids that can be produced endogenously by the human body through glucose synthesis are classified as _____ amino acids.
a.) non-essential b.) dietary c.) essential d.) exogenous
- _____ **106.** The human body is able to produce non-essential amino acids endogenously through the synthesis of _____.
a.) nucleotides b.) glucose c.) fats d.) steroids
- _____ **107.** Out of the 20 standard amino acids, there are _____ non-essential amino acids that the body can synthesize.
a.) 11 b.) 4 c.) 9 d.) 20
- _____ **108.** Which of the following is an amino acid explicitly classified in the lecture notes as a non-essential amino acid?
a.) alanine b.) valine c.) leucine d.) histidine
- _____ **109.** Amino acids that cannot be produced by the body and must be supplied through dietary food intake are _____ amino acids.
a.) endogenous b.) synthetic c.) non-essential d.) essential
- _____ **110.** There are _____ essential amino acids that must be obtained from dietary sources.
a.) 20 b.) 9 c.) 11 d.) 5
- _____ **111.** Essential amino acids must be consumed through dietary sources like fish, poultry, egg, and _____.
a.) starch b.) cellulose c.) sugar d.) meat
- _____ **112.** Which of the following is an amino acid explicitly classified in the lecture notes as an essential amino acid?
a.) valine b.) alanine c.) glycine d.) serine
- _____ **113.** Proteins exhibit structural organization categorized into _____ hierarchical levels.
a.) five b.) three c.) four d.) two
- _____ **114.** The specific, linear sequence of amino acids in a polypeptide chain constitutes a protein's _____ structure.
a.) tertiary b.) primary c.) secondary d.) quaternary
- _____ **115.** A specific biological protein cited in the lecture notes as an example of a defined primary structure is _____.
a.) hemoglobin b.) insulin c.) cellulose d.) DNA polymerase
- _____ **116.** The structural level resulting from the localized bending and coiling of an amino acid chain into pleated sheets or helices is the _____ structure.
a.) quaternary b.) primary c.) secondary d.) tertiary
- _____ **117.** Secondary protein structure results from polypeptide chains bending and coiling into either helices or _____ sheets.
a.) amphipathic b.) globular c.) pleated d.) variable
- _____ **118.** In the study of secondary protein structure, helices and pleated sheets are associated with amino acid derivatives including proline, hydroxyproline, sarcosine, and _____ acid.
a.) pipecolic b.) citric c.) glutamic d.) aspartic
- _____ **119.** Proline, hydroxyproline, pipecolic acid, and sarcosine are specifically noted in the text in relation to _____ protein structure.
a.) primary b.) quaternary c.) secondary d.) tertiary
- _____ **120.** The overall three-dimensional shape of a single folded protein chain is its _____ structure.
a.) primary b.) quaternary c.) tertiary d.) secondary
- _____ **121.** Tertiary protein structure is functionally observed in transport proteins and _____.
a.) monosaccharides b.) enzymes c.) steroids d.) triglycerides
- _____ **122.** Proteins that facilitate movement across membranes and exhibit characteristic three-dimensional tertiary shapes include enzymes and _____ proteins.
a.) fibrous b.) structural c.) transport d.) storage
- _____ **123.** The complex protein structure produced when attractive forces link multiple polypeptide chains (subunits) together is _____ structure.

a.) tertiary b.) secondary c.) quaternary d.) primary

_____ **124.** Quaternary protein structure is distinguished by being comprised of multiple _____ chains.

a.) polysaccharide b.) monosaccharide c.) fatty acid d.) polypeptide

_____ **125.** The oxygen-binding protein in red blood cells that is cited as a classic example of quaternary structure is _____.

a.) collagen b.) hemoglobin c.) insulin d.) myoglobin

_____ **126.** The multi-subunit enzyme responsible for synthesizing new DNA strands, cited as an example of quaternary structure, is _____.

a.) lactase b.) ribonuclease c.) DNA polymerase d.) pepsin

_____ **127.** Transmembrane protein assemblies that regulate electrical potential and ion passage, cited as an example of quaternary structure, are _____.

a.) chitin fibers b.) phospholipids c.) steroid receptors d.) ion channels

_____ **128.** Proteins that act as biological catalysts by speeding up chemical reactions without being consumed in the process are _____.

a.) antibodies b.) hormones c.) enzymes d.) receptors

_____ **129.** A key defining feature of biological enzymes is that they accelerate reactions without being _____ in the process.

a.) consumed b.) synthesized c.) folded d.) activated

_____ **130.** The specific chemical reactant molecule upon which an enzyme acts is the _____.

a.) substrate b.) product c.) cofactor d.) coenzyme

_____ **131.** The specialized region or binding pocket on an enzyme protein where the substrate binds is the _____ site.

a.) active b.) regulatory c.) variable d.) allosteric

_____ **132.** When a substrate binds to the active site of an enzyme, they temporarily form an _____ complex.

a.) enzyme–inhibitor b.) ionic–covalent c.) enzyme–substrate d.) active–allosteric

_____ **133.** Following catalytic conversion within the enzyme-substrate complex, the substrate is transformed into _____ that leave the active site.

a.) products b.) monomers c.) cofactors d.) reactants

_____ **134.** Nonprotein chemical substances required by certain enzymes to convert substrate to product are called _____.

a.) cofactors b.) monomers c.) polymers d.) inhibitors

_____ **135.** Organic cofactors, such as dietary vitamins, that assist enzymes in performing catalytic functions are specifically called _____.

a.) coenzymes b.) substrates c.) amphipaths d.) isotopes

_____ **136.** Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are polymers composed of repeating subunits called _____.

a.) monosaccharides b.) amino acids c.) fatty acids d.) nucleotides

_____ **137.** A nucleotide consists of a five-carbon sugar, a phosphate group, and one of five _____ bases.

a.) carbon-saturated b.) nitrogen-containing c.) sulfur-rich d.) oxygen-free

_____ **138.** The pentose sugar present in every nucleotide molecule contains _____ carbon atoms.

a.) three b.) six c.) five d.) seven

_____ **139.** Stretches of DNA that determine the sequence of amino acids that will assemble into a protein are called a _____.

a.) nucleolus b.) ribosome c.) gene d.) codon

_____ **140.** A gene consists of a stretch of DNA that encodes instructions to determine the sequence of _____ in a protein.

a.) monosaccharides b.) fatty acids c.) amino acids d.) nucleotides

- _____ **141.** DNA and RNA are biochemically distinguished from each other by their structure, nitrogen-containing bases, and _____ component.
a.) sugar b.) lipid c.) protein d.) phosphate
- _____ **142.** In terms of molecular architecture, DNA is characteristically a _____-stranded molecule.
a.) branched b.) triple c.) single d.) double
- _____ **143.** The five-carbon pentose sugar component found in DNA nucleotides is _____.
a.) fructose b.) ribose c.) glucose d.) deoxyribose
- _____ **144.** The four nitrogen-containing bases found in DNA nucleotides are adenine, guanine, cytosine, and _____.
a.) ribose b.) thymine c.) histidine d.) uracil
- _____ **145.** In contrast to DNA, RNA is characteristically a _____-stranded molecule.
a.) single b.) circular c.) double d.) triple
- _____ **146.** The five-carbon pentose sugar component found in RNA nucleotides is _____.
a.) ribose b.) galactose c.) glucose d.) deoxyribose
- _____ **147.** The four nitrogen-containing bases found in RNA nucleotides are adenine, guanine, cytosine, and _____.
a.) alanine b.) uracil c.) valine d.) thymine
- _____ **148.** The nitrogenous base found in RNA that substitutes for thymine and pairs with adenine is _____.
a.) guanine b.) thymine c.) uracil d.) cytosine
- _____ **149.** In human cells, RNA molecules serve as structural and catalytic components of protein-making organelles called _____.
a.) vacuoles b.) ribosomes c.) lysosomes d.) mitochondria
- _____ **150.** During protein synthesis, transfer RNA (tRNA) molecules act specifically as transfer agents of _____.
a.) glucose units b.) amino acids c.) fatty acids d.) phosphate ions
- _____ **151.** During protein synthesis, messenger RNA (mRNA) functions as the carrier of genetic _____ from DNA to the ribosome.
a.) radiation b.) glucose c.) enzymes d.) information
- _____ **152.** In cell biology, RNA plays three primary roles in protein synthesis: as messengers of information, transfer agents of amino acids, and components of _____.
a.) plasma membranes b.) ribosomes c.) chromosomes d.) centrioles

COMPLETE ANSWER KEY & EXPLANATIONS

Module 1: The Scientific Method & Basic Chemistry — Reference Key with Concept Explanations

Item	Key	Correct Answer	Curriculum Explanation & Reference
1	C	scientific method	[Scientific Method Definition] The scientific method is defined as a systematic approach to investigating the natural world through logical inquiry, rigorous observation, and evidence-based conclusion.
2	B	question	[Phase 1: Observation & Question] Phase 1 involves making careful observations of the environment, which in turn prompts a specific question informed by existing knowledge.
3	C	environment	[Phase 1: Environmental Observation] Observation of the natural environment provides the empirical foundation upon which scientific questions are formulated.
4	A	hypothesis	[Phase 2: Hypothesis Formulation] Phase 2 requires developing a testable hypothesis, which represents a plausible explanation or answer to the question asked.
5	C	statement	[Phase 2: Hypothesis Format] A hypothesis must be framed as a definitive declarative statement (e.g., 'Daily exposure to the sun leads to increased levels of happiness') rather than a question.
6	C	hypothesis	[Phase 2: Hypothesis Example] The proposition 'Daily exposure to the sun leads to increased levels of happiness' is an example of a testable declarative statement formulating a hypothesis.
7	D	false	[Phase 2: Testability & Falsifiability] For a hypothesis to be scientifically valid, it must be falsifiable—meaning there must be conceivable empirical observations that could prove it false.
8	A	true	[Phase 2: Provisional Nature of Science] A hypothesis can be supported repeatedly, but in empirical science, it can never be proven absolutely true beyond all future revision.
9	B	prediction	[Phase 3: Prediction] A prediction forecasts the specific expected outcome of an experiment if the hypothesis holds true.
10	C	test	[Phase 3: Testing Predictions] The prediction dictates what specific empirical data and observations must be collected to test the validity of the hypothesis.
11	C	controlled	[Phase 3: Controlled Experiment] A controlled experiment isolates the effect of an independent variable by maintaining identical baseline conditions across groups.
12	D	control	[Phase 3: Control Group] Controlled experiments require both a control group (baseline reference) and an experimental group (treatment cohort).
13	B	variable	[Phase 3: Variable Distinction] The single factor that differs between

			the control and experimental groups is the variable.
14	C	variable	[Phase 3: Experimental Variable] The variable is the distinct factor being tested to determine its effect on the experimental system.
15	B	conclusion	[Phase 4: Conclusion] Phase 4 culminates in drawing conclusions based on the systematic analysis and interpretation of empirical results.
16	B	data	[Phase 4: Data Interpretation] Scientific conclusions must be derived strictly from the objective interpretation of empirical data gathered during testing.
17	D	statistical significance	[Phase 4: Statistical Significance] Statistical significance ensures mathematically that observed experimental results are unlikely to have arisen by random chance.
18	C	matter	[Matter Definition] Matter is defined as any physical substance that has mass and occupies volume in space.
19	C	atoms	[Atoms] Atoms are the indivisible basic units of matter that retain chemical identity and cannot be chemically cleaved.
20	A	neutrons	[Subatomic Particles] Atoms consist of three fundamental subatomic particles: protons, neutrons, and electrons.
21	B	proton	[Protons] Protons are positively charged subatomic particles that reside in the nucleus of an atom.
22	D	neutron	[Neutrons] Neutrons are uncharged, neutral subatomic particles present alongside protons in the atomic nucleus.
23	A	electron	[Electrons] Electrons are negatively charged subatomic particles that occupy orbitals surrounding the atomic nucleus.
24	C	element	[Elements] An element is a pure substance consisting entirely of one specific species of atom.
25	A	100	[Known Elements] Modern science recognizes more than 100 distinct chemical elements.
26	B	20	[Elements in Human Body] Approximately 20 chemical elements are biologically utilized and found within the human body.
27	D	hydrogen	[Major Elements of Human Body] Carbon, Oxygen, Hydrogen, and Nitrogen (COHN) constitute roughly 96% of human body mass.
28	D	one or two-letter	[Chemical Symbols] Elements are standardly represented by a one or two-letter symbol derived from their English or Latin names (e.g., C, O, Na, He).
29	B	atomic mass unit	[Atomic Mass Unit] Atomic weight or mass is measured in atomic mass

units (amu), also known as Daltons.

30	A	neutrons	[Determinants of Atomic Weight] Because electrons have negligible mass, an atom's atomic weight is determined almost entirely by the sum of its protons and neutrons.
31	C	isotopes	[Isotopes] Isotopes are variants of a specific chemical element having identical proton counts but different numbers of neutrons.
32	D	radioisotopes	[Radioisotopes] Radioisotopes are radioactive isotopes that decay spontaneously, emitting energetic radiation or particles.
33	A	radiation	[Radioactive Decay Emission] The breakdown of unstable radioisotopes releases energy in the form of radiation or subatomic particles.
34	A	compound	[Chemical Compounds] A compound is a substance formed when two or more elements combine chemically, acquiring completely novel chemical and physical properties.
35	B	bonds	[Chemical Bonds] Chemical bonds are the attractive forces that link atoms together to create stable molecules and compounds.
36	D	covalent	[Bond Categories] Chemical bonds in biological systems are primarily categorized into covalent bonds and ionic bonds.
37	B	covalent	[Covalent Bonds] Covalent bonds form when atoms mutually share pairs of electrons in their outermost valence shells.
38	A	outer shells	[Covalent Electron Sharing] Covalent electron sharing occurs between the valence electrons located in the outermost electron shells (cells) of the interacting atoms.
39	A	ionic	[Ionic Bonds] Ionic bonds result from the transfer of valence electrons, producing cations and anions whose opposite electrical charges attract.
40	D	oppositely	[Ionic Bond Attraction] Ionic bonding occurs due to the electrostatic attraction between positively charged ions (cations) and negatively charged ions (anions).
41	A	solvent	[Water as Solvent] Water's polar molecular structure enables it to dissolve a wide array of polar and ionic solutes, functioning as the body's primary solvent.
42	A	heat capacity	[Water Heat Capacity] High heat capacity enables water to absorb and store thermal energy, stabilizing biological temperatures against fluctuations.
43	D	vaporization	[Heat of Vaporization] Water's high heat of vaporization requires substantial thermal energy to convert liquid water into vapor, shedding body heat during sweating.

44	A	polarity	[Water Polarity] The polarity of the water molecule and its resulting ability to form hydrogen bonds account for water's unique thermal and chemical properties.
45	C	partial	[Polarity Definition] Polarity refers to the unequal distribution of shared electrons, causing partial negative charges on oxygen and partial positive charges on hydrogens.
46	C	hydrogen	[Hydrogen Bonds] A hydrogen bond is an intermolecular attraction between the slightly positive hydrogen of one polar molecule and the slightly negative atom of another.
47	A	positive	[Hydrogen Bond Charges] Hydrogen bonds arise specifically from the electrical attraction between the slightly positive hydrogen atom and a slightly negative oxygen atom.
48	A	macromolecules	[Macromolecules] Macromolecules are giant organic molecules constructed from thousands of covalently bonded atoms.
49	D	polymers	[Polymers] A polymer is a large macromolecule composed of repeated, linked chemical units called monomers.
50	D	monomers	[Monomers] Monomers are the fundamental sub-units that undergo polymerization to assemble large polymers.
51	D	dehydration	[Dehydration Synthesis] Dehydration synthesis (condensation) reactions link monomers into polymers through the removal of water.
52	C	hydrolysis	[Hydrolysis Reactions] Hydrolysis reactions cleave chemical bonds within polymers by adding water molecules, splitting them back into monomers.
53	B	hydrolysis	[Dehydration vs Hydrolysis] Dehydration builds polymers by joining subunits, whereas hydrolysis breaks polymers down into smaller components.
54	D	fats	[Four Major Biomolecules] Living organisms are built from four classes of biological macromolecules: carbohydrates, proteins, fats (lipids), and nucleic acids.
55	C	carbohydrates	[Carbohydrates Overview] Carbohydrates are composed of C, H, and O and function as the body's premier source of cellular fuel and metabolic energy.
56	D	sugars	[Carbohydrate Examples] Sugars and starches represent the primary nutritional forms of carbohydrate biomolecules.
57	A	oligosaccharides	[Carbohydrate Classification] Carbohydrates are grouped into monosaccharides (single sugars), oligosaccharides (short chains), and polysaccharides (complex chains).
58	B	monosaccharides	[Monosaccharides] Monosaccharides are the basic monomeric sugar

units, also called simple sugars.

59	B	seven	[Monosaccharide Carbon Range] Monosaccharides are classified by their carbon backbone length, which spans from 3 to 7 carbon atoms.
60	A	hexoses	[Hexose Sugars] Hexoses are six-carbon monosaccharides (C ₆ H ₁₂ O ₆), including glucose, fructose, and galactose.
61	C	glucose	[Glucose] Glucose is the six-carbon hexose that serves as the central fuel molecule for cellular respiration in human tissues.
62	A	fructose	[Fructose] Fructose is a six-carbon hexose found naturally in fruits, honey, and vegetables.
63	C	galactose	[Galactose] Galactose is a six-carbon hexose sugar abundant in milk and dairy products as part of lactose.
64	D	ribose	[Ribose] Ribose is a 5-carbon pentose sugar that forms an integral component of RNA and ATP.
65	B	pentose	[Pentose Definition] Pentoses are monosaccharides containing exactly five carbon atoms.
66	A	oligosaccharides	[Oligosaccharides] Oligosaccharides are short polymers consisting of a few monosaccharide units linked by glycosidic bonds.
67	B	subunits	[Oligosaccharide Classification] Oligosaccharides are classified by the exact number of monosaccharide subunits in their chain (e.g., disaccharides, trisaccharides).
68	C	disaccharides	[Disaccharides] A disaccharide is a carbohydrate formed by covalently linking two monosaccharide units.
69	D	sucrose	[Sucrose] Sucrose (table sugar) is a disaccharide made of one glucose molecule bonded to one fructose molecule.
70	A	polysaccharides	[Polysaccharides] Polysaccharides are large, complex macromolecules composed of hundreds or thousands of monosaccharide units.
71	B	storage	[Polysaccharide Functional Classes] Polysaccharides are functionally categorized as either storage polysaccharides (energy reserves) or structural polysaccharides (support).
72	D	storage	[Storage Polysaccharides] Storage polysaccharides sequester glucose molecules in compact forms for future metabolic breakdown.
73	B	starch	[Starch] Starch is the primary energy storage polysaccharide synthesized by plants.
74	A	glycogen	[Glycogen] Glycogen is the major glucose storage polymer utilized by animals and humans.

75	A	structural	[Structural Polysaccharides] Structural polysaccharides form tough fibrous networks that give cells and organisms structural rigidity.
76	B	cellulose	[Cellulose] Cellulose is an unbranched polymer of glucose that forms plant cell walls and fibers in bamboo, jute, and cotton.
77	B	cellulose	[Cellulose Examples] Bamboo, jute, and cotton are natural fibrous materials composed predominantly of cellulose.
78	B	lipids	[Lipids Overview] Lipids (fats) are hydrophobic organic compounds that are insoluble in water because of their nonpolar hydrocarbon chains.
79	B	triglycerides	[Three Lipid Types] The three major classes of biologically important lipids are triglycerides, phospholipids, and steroids.
80	A	fatty acids	[Triglyceride Composition] Triglycerides are formed by esterifying three fatty acid chains to a single glycerol backbone.
81	A	COOH	[Fatty Acid Functional Group] Fatty acids consist of a long hydrocarbon tail capped at one end by a carboxyl group (-COOH).
82	D	hydrogens	[Fatty Acid Hydrocarbon Chain] Fatty acid tails consist of long hydrocarbon chains where carbon atoms are bonded to hydrogen atoms.
83	C	double bonds	[Fatty Acid Classification] Fatty acids are classified as saturated or unsaturated depending on whether double bonds are present between carbon atoms.
84	C	saturated	[Saturated Fatty Acids] Saturated fatty acids contain only single bonds between carbons, holding the maximum possible number of hydrogen atoms.
85	A	unsaturated	[Unsaturated Fatty Acids] Unsaturated fatty acids contain one or more carbon-carbon double bonds, creating bends or kinks in the tail.
86	D	oils	[Triglyceride Examples] Triglycerides manifest dietary-wise as solid fats (usually animal-derived) and liquid oils (usually plant-derived).
87	A	twice	[Energy Density of Triglycerides] Lipids store roughly 9 kcal/g, which is more than twice the caloric energy density of carbohydrates or proteins (4 kcal/g).
88	B	triglyceride	[Most Common Body Fat] Triglycerides represent the predominant form of stored body fat in human adipose tissue.
89	A	oils	[Dietary Sources of Triglycerides] Dietary triglycerides originate from foods rich in fats, notably butter, oils, margarine, and animal fats.
90	B	two	[Phospholipid Composition] Phospholipids consist of a glycerol backbone bonded to two fatty acid tails and one phosphate head.
91	B	variable	[Phospholipid Variable Group] Phospholipids typically have a variable

			organic group (such as choline or serine) attached to their phosphate head.
92	A	amphipathic	[Amphipathic Nature] Amphipathic molecules feature dual chemical affinities, having both polar/charged hydrophilic and nonpolar hydrophobic domains.
93	C	plasma	[Plasma Membrane Phospholipids] The amphipathic nature of phospholipids drives them to self-assemble into the lipid bilayer of cellular plasma membranes.
94	D	four	[Steroid Ring Structure] Steroids are characterized by a fused carbon skeleton consisting of four interconnected rings (three six-carbon rings and one five-carbon ring).
95	D	variable	[Steroid Variable Molecules] Different steroids vary according to the specific variable chemical groups attached to their core four carbon rings.
96	B	cholesterol	[Cholesterol] Cholesterol stabilizes the fluidity of cell membranes and acts as the biosynthetic foundation for steroid hormones.
97	D	estrogen	[Steroid Hormones] Steroid hormones, including estrogen, progesterone, and testosterone, are synthesized biochemically from cholesterol.
98	C	proteins	[Proteins Overview] Proteins are complex macromolecules assembled from chains of 20 standard amino acid monomers.
99	B	amino acid	[Amino Acids] Amino acids are the building blocks that polymerize to form polypeptide chains and functional proteins.
100	A	carbon	[Central Carbon Atom] An amino acid features a central alpha-carbon (C) atom to which all its functional groups attach.
101	C	amino	[Amino Group] The amino group (-NH ₂) is the basic, nitrogen-containing component of an amino acid.
102	D	carboxyl	[Carboxyl Group] The carboxyl group (-COOH) is the acidic chemical moiety bound to the alpha-carbon of amino acids.
103	A	side chain	[Side Chain (R Group)] The side chain (R-group) is the unique variable group that confers distinctive chemical properties to each of the 20 amino acids.
104	B	body	[Amino Acid Classification Basis] Amino acids are categorized based on whether the human body possesses the metabolic pathways to synthesize them.
105	A	non-essential	[Non-Essential Amino Acids] Non-essential amino acids are those that the human body can synthesize internally, so they are not strictly required in the diet.

106	B	glucose	[Glucose Synthesis of Amino Acids] Non-essential amino acids are biosynthesized using carbon skeletons derived from glucose metabolism.
107	A	11	[Count of Non-Essential Amino Acids] There are 11 non-essential amino acids (alanine, arginine, asparagine, aspartic acid, cysteine, glutamic acid, glutamine, glycine, proline, serine, and tyrosine).
108	A	alanine	[Non-Essential Amino Acid Example] Alanine is one of the 11 non-essential amino acids synthesized by the human body.
109	D	essential	[Essential Amino Acids] Essential amino acids cannot be synthesized by human cells and must be obtained from food.
110	B	9	[Count of Essential Amino Acids] There are 9 essential amino acids: histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.
111	D	meat	[Dietary Sources of Essential Amino Acids] Complete dietary protein sources rich in essential amino acids include meat, poultry, fish, and eggs.
112	A	valine	[Essential Amino Acid Example] Valine is one of the 9 essential amino acids that must be acquired through the diet.
113	C	four	[Four Levels of Protein Structure] Protein structure is organized into four levels: primary, secondary, tertiary, and quaternary.
114	B	primary	[Primary Protein Structure] Primary structure is the unique linear sequence of covalently linked amino acids in a polypeptide.
115	B	insulin	[Insulin Primary Structure] Insulin is the classic hormone referenced for its precisely sequenced primary chain of amino acids.
116	C	secondary	[Secondary Protein Structure] Secondary structure results from hydrogen bonding along the polypeptide backbone, creating alpha-helices and beta-pleated sheets.
117	C	pleated	[Pleated Sheets] The two predominant secondary conformations are alpha-helices and beta-pleated sheets.
118	A	pipecolic	[Pipecolic Acid in Secondary Structure] The lecture text explicitly associates secondary structural bending and coiling with proline, hydroxyproline, pipecolic acid, and sarcosine.
119	C	secondary	[Secondary Structure Amino Acids] These specific imino and amino acid residues influence the formation of secondary structural helices and sheets.
120	C	tertiary	[Tertiary Protein Structure] Tertiary structure represents the complete 3D spatial conformation of a folded polypeptide chain.
121	B	enzymes	[Tertiary Structure Examples] Enzymes and membrane transport

			proteins rely on precise tertiary three-dimensional conformations to perform their functions.
122	C	transport	[Transport Proteins] Transport proteins possess specific 3D tertiary conformations allowing them to shuttle solutes across membranes.
123	C	quaternary	[Quaternary Protein Structure] Quaternary structure is formed when two or more polypeptide subunits associate into a multi-subunit functional complex.
124	D	polypeptide	[Polypeptide Subunits] Quaternary proteins consist of two or more independent polypeptide chains joined by non-covalent or covalent bonds.
125	B	hemoglobin	[Hemoglobin Quaternary Structure] Hemoglobin is composed of four polypeptide subunits (two alpha and two beta chains), exhibiting quaternary structure.
126	C	DNA polymerase	[DNA Polymerase Quaternary Structure] DNA polymerase is a complex multi-subunit holoenzyme with quaternary structure.
127	D	ion channels	[Ion Channels Quaternary Structure] Ion channels are multi-subunit protein complexes that span biological membranes, exemplifying quaternary architecture.
128	C	enzymes	[Enzymes as Biological Catalysts] Enzymes are protein catalysts that accelerate biochemical reactions by lowering activation energy, remaining unaltered after the reaction.
129	A	consumed	[Enzymatic Catalysis] Catalysts facilitate chemical reactions without being consumed or permanently altered in the process.
130	A	substrate	[Enzyme Substrate] The substrate is the specific reactant molecule that binds to an enzyme to undergo biochemical conversion.
131	A	active	[Enzyme Active Site] The active site is the specialized catalytic pocket on an enzyme where substrate molecules bind.
132	C	enzyme–substrate	[Enzyme–Substrate Complex] Substrate binding to the active site produces a temporary enzyme-substrate intermediate complex.
133	A	products	[Enzymatic Products] Once catalysis is complete, newly formed products detach and release from the enzyme's active site.
134	A	cofactors	[Cofactors] Cofactors are nonprotein helper components (such as metal ions) required by enzymes for catalytic activity.
135	A	coenzymes	[Coenzymes] Coenzymes are organic cofactors (often derived from vitamins) that assist enzymes in transferring chemical groups.
136	D	nucleotides	[Nucleic Acids & Nucleotides] DNA and RNA are nucleic acid polymers

			composed of nucleotide monomer building blocks.
137	B	nitrogen-containing	[Nucleotide Components] Every nucleotide is built from three components: a five-carbon pentose sugar, a phosphate group, and a nitrogenous base.
138	C	five	[Nucleotide Pentose Sugar] Nucleotides always contain a 5-carbon pentose sugar (deoxyribose in DNA, ribose in RNA).
139	C	gene	[Genes] A gene is a functional segment of DNA that encodes the precise amino acid sequence of a polypeptide chain.
140	C	amino acids	[Genetic Code for Amino Acids] Genes encode genetic information that translates into the primary sequence of amino acids during protein synthesis.
141	A	sugar	[DNA vs RNA Distinctions] DNA and RNA are distinguished by their strandedness (structure), their pentose sugar (deoxyribose vs ribose), and their bases.
142	D	double	[DNA Double-Stranded Structure] DNA consists of two antiparallel nucleotide strands wound into a double helix.
143	D	deoxyribose	[Deoxyribose] DNA contains deoxyribose sugar, which lacks an oxygen atom at the 2' carbon position compared to ribose.
144	B	thymine	[DNA Bases (Thymine)] DNA contains four nitrogenous bases: Adenine (A), Guanine (G), Cytosine (C), and Thymine (T).
145	A	single	[RNA Single-Stranded Structure] RNA molecules typically exist as single-stranded polynucleotide chains.
146	A	ribose	[Ribose in RNA] RNA contains ribose sugar, possessing a 2'-hydroxyl group.
147	B	uracil	[RNA Bases (Uracil)] RNA contains Adenine (A), Guanine (G), Cytosine (C), and Uracil (U), lacking thymine.
148	C	uracil	[Uracil Replaces Thymine] In RNA, uracil replaces thymine and forms complementary base pairs with adenine.
149	B	ribosomes	[RNA in Ribosomes] Ribosomal RNA (rRNA) combines with proteins to form the structure of ribosomes, the protein synthesis machines.
150	B	amino acids	[tRNA Transfer of Amino Acids] tRNA transfers specific amino acids to the growing polypeptide chain at the ribosome.
151	D	information	[mRNA Messenger of Information] mRNA acts as the messenger carrying transcribed genetic information from nuclear DNA to cytoplasmic ribosomes.
152	B	ribosomes	[Three Roles of RNA] The three core functions of RNA in protein

synthesis are: mRNA (information messenger), tRNA (amino acid transfer), and rRNA (ribosome component).
